Analyzing the Relationship Between SSDI Recipients and University Enrollment Trends (2005-2024)

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Introduction

This research paper uses two data sets that range from 2005-2024. The first data set comes from Western Michigan University (WMU) and includes their total student enrollment each year from 2005-2024. The second data set also ranges from 2005-2024 and shows the number of disabled workers entitled to SSDI benefits each year. The reason why these two specific data sets were chosen is that they both track information over an extended period of time. Correlating the two may allow us to understand if there are any common trends between the two like the amount of people in the work force or not going to college.

For this assignment, WMU enrollment will be our Y or Dependent variable and the number of disabled workers entitled to SSDI benefits will be our X or Independent variable. Both are ratio level data and will use time (year) as the data point for each set. Each data set uses the same time frame which meets the qualifications for regression and correlation. It is hoped that this analysis will reveal whether declining university enrollment and fluctuating SSDI participation share any meaningful relationship, potentially reflecting broader economic or demographic trends affecting both higher education and the workforce simultaneously.

The hypotheses for this study are:

- **H_0:** $\beta_1 = 0$ (There is no significant linear relationship between SSDI recipients and WMU enrollment)

- **H_1:** $\beta_1 \neq 0$ (There is a significant linear relationship between SSDI recipients and WMU enrollment)

Research Question: Is there a significant relationship between disabled workers entitled to SSDI benefits and WMU enrollment?

Table 1.0. Raw Data.

Data Name: WMU Total Enrollment			**Data Name: Disabled Workers Receiving SSDI**	
Year	**Y**		**Year**	**X**
2005	26,239		2005	6,519,001
2006	24,841		2006	6,806,918
2007	24,433		2007	7,098,723
2008	24,818		2008	7,426,691
2009	24,576		2009	7,788,013
2010	25,045		2010	8,203,951
2011	25,086		2011	8,575,544
2012	24,598		2012	8,826,591
2013	24,294		2013	8,940,950
2014	23,914		2014	8,954,518
2015	23,556		2015	8,909,430
2016	23,252		2016	8,808,736
2017	22,894		2017	8,695,475
2018	22,562		2018	8,537,332
2019	21,470		2019	8,378,374
2020	20,490		2020	8,151,016
2021	19,038		2021	7,877,129
2022	17,835		2022	7,604,098
2023	17,559		2023	7,365,987
2024	17,605		2024	7,231,147

Frequency/Percentage Distributions
Graph of Each Frequency Distribution

Table 2.0. Frequency/Percentage Distribution WMU Enrollment ’05-‘24

Class	Count	Percent
17,500-19,699	4	20%
19,700-21,899	2	10%
21,900-24,099	5	25%
24,100-26,299	9	45%
Total	**20**	**100%**
Class Width	2200	
Simple Method	2,199	
Boundary	0.5	
Boundary Method	2200	

Figure 2.0. Frequency Distribution Graph WMU Enrollment ’05-‘24

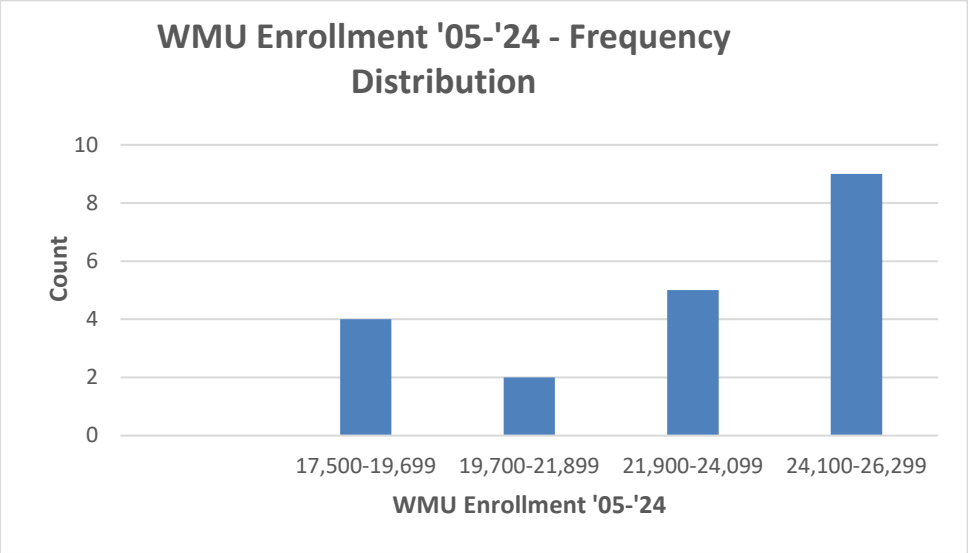
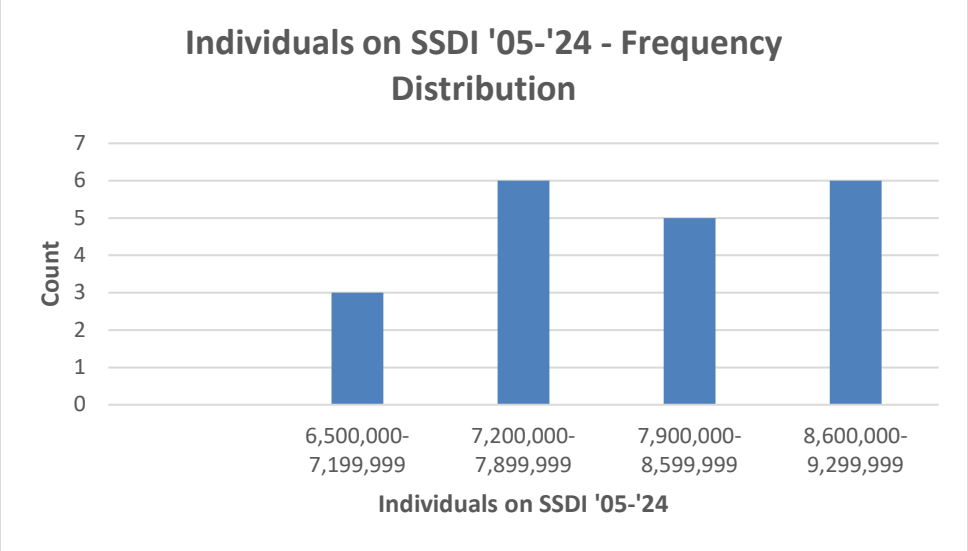


Table 2.1. Frequency/Percentage Distribution Disabled Workers on SSDI ’05-‘24

Class	Count	Percent
6,500,000-7,199,999	3	15%

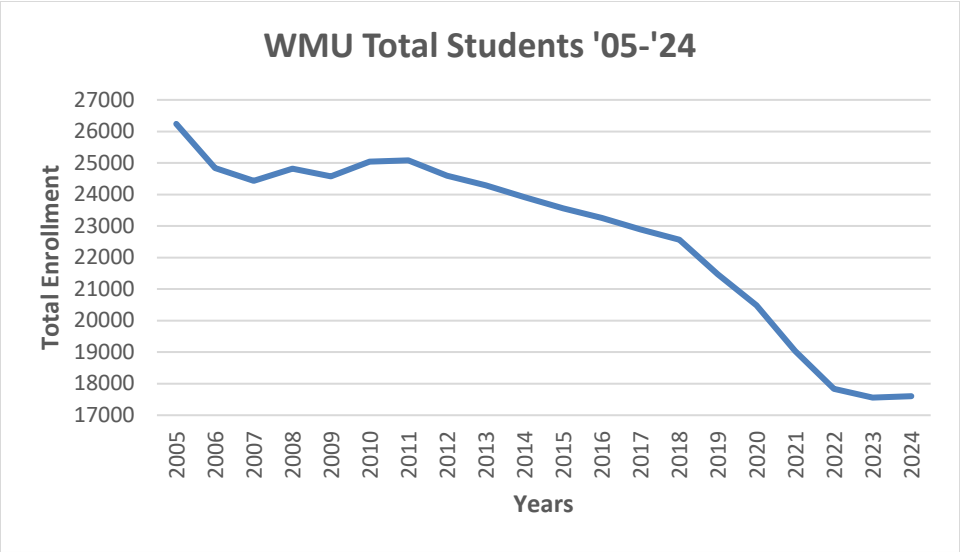
7,200,000-7,899,999	6	30%
7,900,000-8,599,999	5	25%
8,600,000-9,299,999	6	30%
Total	**20**	**100%**
Class Width	700,000	
Simple Method	699,999	
Boundary	0.5	
Boundary Method	700000	

Figure 2.1. Frequency Distribution Graph Disabled Workers on SSDI ’05-‘24



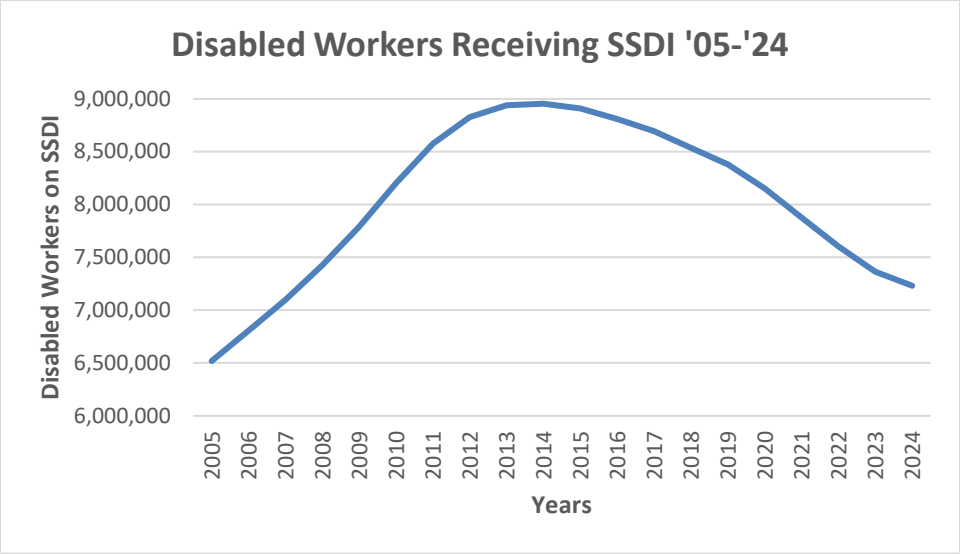
Graphs of Raw Data

Figure 3.0. WMU Total Students ’05-‘24



The descriptive statistics for WMU enrollment shows that the mean enrollment is 22,705.25 and the median enrollment is 23,735.00. Since the mean is less than the median we can assume enrollment amounts have been dragged down slightly in the more recent years. The range and standard deviation are pretty large (range = 8,680 and standard deviation = 2,751.8797) due to the fact that enrollment has steadily decreased each year. Since this is a set of data over time, there should not be a mode.

Figure 3.1. Disabled Workers Receiving SSDI ’05-‘24



The SSDI dataset has a mean value of 8,034,981 with a median value of 8,177,484. This means that the data is very slightly skewed. The variance and standard deviation are large due to the fact that the standard deviation is 768,295.6479. This makes sense because the number of SSDI recipients have changed greatly over time. There is a huge upward trend and then a slower decrease.

Descriptive Statistics

Table 4.0. Descriptive Statistics Data for WMU Total Enrollment ’05-‘24.

WMU Total Enrollment ’05 - ‘24	
Mean	22,705.2500
Median	23,735
Mode	N/A
Range	8,680
Midrange	21,899
Variance	7,572,841.6711
Standard Deviation	2,751.8797

WMU Enrollment has an overall negative distribution. We know this because the mean is less than the median. Because enrollment is steadily declining over time this is pushing the mean lower and creating a negative distribution. Enrollment is skewed negatively because there are enrollment amounts that are lower than recent years. WMU Enrollment has a large standard deviation and variance due to enrollment steadily declining from 26,239 students in 2005 to 17,605 students in 2024. There is no mode, which is typical for continuous data measured over time. The midrange of 21,899 falls slightly below the mean, further supporting the presence of a downward trend in enrollment over the 20-year period.

Table 4.1. Descriptive Statistics Data for Disabled Workers on SSDI ’05-‘24.

Disabled Workers on SSDI ’05 - ‘24	
Mean	8,034,981.2000
Median	8,177,484
Mode	N/A
Range	2,435,517
Midrange	7,736,759.50
Variance	590,278,202,580
Standard Deviation	768,295.6479

The descriptive statistics of SSDI recipients present a somewhat skewed distribution since the study period is not linear. The mean of 8,034,981 and median of 8,177,484 are fairly close together which suggests the distribution is only slightly skewed. However, due to the large range of values during the 20 years studied (captured by a large increase in SSDI recipients until 2014 and a slow decrease thereafter), the standard deviation of 768,295.6479 and variance of 590,278,202,580 are quite high. This is also shown by the range of 2,435,517. There should be no mode since this is time-series data. The midrange of 7,736,759.50 is less than both the mean and median which makes sense since the values at the beginning and end of the study period are less than the majority of other values.

Confidence Intervals

Table 5.0. Confidence Interval Data WMU Enrollment ’05-‘24.

WMU Enrollment ’05-‘24	Data
Step 1 Standard Error	615.3390
Step 2 t value	2.093
Step 3 Interval Estimate	21,417.3455 <Pop. Mean< 23,993.1545

The confidence interval when estimating the mean using a sample size of 20 years of enrollment data, with a standard error of 615.3390 and t-value of 2.093, is between 21,417.3455 and 23,993.1545 students enrolled at WMU. We can say with 95% certainty that the true population mean lies somewhere in that range. Note how wide of a range this is. Also note how the most recent enrollment numbers we have (17,559 students in 2023 and 17,605 students in 2024) would not fall within that range if enrollment continues to trend downward as it has recently.

Table 5.1. Confidence Interval Data Disabled Workers on SSDI ’05-‘24.

Workers on SSDI ’05-‘24	Data
Step 1 Standard Error	171,796.1296
Step 2 t value	2.093
Step 3 Interval Estimate	7,675,411.901 <Pop. Mean< 8,394,550

Considering a sample of twenty years of SSDI data with a standard error of 171,796.1296 and a t-score of 2.093, we are 95% confident that the interval from 7,675,411.9010 to 8,394,550.4990 captures the true population mean of disabled workers collecting SSDI. This interval covers close to 720,000 recipients and is largely impacted by the tremendous growth in SSDI participation up until 2014 and subsequent slow downturn. This interval range confirms our previous insights gathered from the

standard deviation and variance. We can see that the number of SSDI recipients varied greatly and was affected by something other than stability from one year to the next. (Social Security Administration, n.d.)

Statistical Testing

Step 1 - **H0:** $\beta_1 = 0$ (There is no significant linear relationship between SSDI recipients and WMU enrollment.)

H1: $\beta_1 \neq 0$ (There is a significant linear relationship between SSDI recipients and WMU enrollment.)

Step 2 - The level of significance is $\alpha = 0.05$. Reject H_0 if the p-value < 0.05.

Step 3

Table 6.0. Regression/Correlation Summary Output.

Correlation Coefficient	.1281
Coefficient of Determination	.0164
Standard Error	2804.0009
P-value	.5905
Regression Equation	$\hat{y}$ = 19019.10 + 0.000459x

Step 4 - If $p > 0.05 \rightarrow$ fail to reject H_0

Result: $p > 0.05$, fail to reject H_0

Step 5 - Since the p-value is greater than 0.05, we fail to reject the null hypothesis.

There is insufficient evidence to conclude that a significant linear relationship exists between SSDI recipients and WMU enrollment.

Conclusions/Summary

The goal of this project was to see if SSDI recipients were significantly related to WMU enrollment through time. After conducting regression analysis and finding our correlation coefficient, we do not have enough statistical evidence to conclude that there is a relationship between SSDI Recipients and WMU Enrollment. Although both variables change over time, WMU enrollment generally declines while SSDI rises through 2014 and then declines, and the statistical relationship between them is not significant. The correlation coefficient is very low at $r = 0.1281$, and the p-value is very high at $p = 0.5905$. This means that there may be some relationship, but it is not statistically significant.

The primary reason we did not find a relationship is that these two variables change based on different attributes. Factors that affect university enrollment trends include demographics, tuition price, school policy, and other higher education trends (Diez et al., 2019). Factors that affect SSDI participants change based on the job market, disability care, and the qualification criteria that are set by the federal government (Social Security Administration, n.d.). Since both of these variables change based on different settings, there will not be a relationship with one another.

Another reason we were unable to find a relationship is because we are comparing one university to a national data set. Since one is at a smaller scale than the other, we are unable to find meaningful relationships. There are always going to be

other external factors that are affecting both of these variables such as economy, population, and changes in public policy.

The main thing we learned from this assignment is that just because two things can be correlated, it does not mean that one causes another (Diez et al., 2019). SSDI participants would not affect university enrollment just because they are decreasing or increasing at similar rates.

This project was good at showing that you cannot always assume that there is a relationship between two variables just because you can see a trend when the data is plotted. Statistical testing is important to help you come to that conclusion.

References

Diez, D. M., Barr, C. D., & Çetinkaya-Rundel, M. (2019). *OpenIntro statistics* (4th ed.). OpenIntro.

Common data set. Common Data Set | Institutional Research | Western Michigan University. (n.d.). https://wmich.edu/institutionalresearch/commondataset

Social Security Administration. The United States Social Security Administration. (n.d.). https://www.ssa.gov/policy/docs/statcomps/di_asr/

Appendix

Appendix A – WMU Total Enrollment ’05-’24 Descriptive Statistics

Mean	22705.2500
Median	23735.0000
Mode	#N/A
MidRange	21899.0000
Range	8680
Population Variance	7194199.5875
Sample Variance	7572841.6711
Population Standard Deviation	2682.2005
Sample Standard Deviation	2751.8797

Column1	
Mean	22705.25
Standard Error	615.3390
Median	23735
Mode	#N/A
Standard Deviation	2751.88
Sample Variance	7572842
Kurtosis	-0.45327
Skewness	-0.9314
Range	8680
Minimum	17559
Maximum	26239
Sum	454105
Count	20

Appendix B – Disabled Workers on SSDI ’05-‘24

Mean	8,034,981.2000
Median	8,177,483.5000
Mode	#N/A
MidRange	7736759.5000
Range	2,435,517.0000
Population Variance	560,764,292,453.8600
Sample Variance	590278202583.0110
Population Standard Deviation	748841.9676
Sample Standard Deviation	768295.6479

Column1	
Mean	8034981.2000
Standard Error	171796.1296
Median	8177483.5000
Mode	#N/A
Standard Deviation	768295.6479
Sample Variance	590278202583.01100
Kurtosis	-1.0104
Skewness	-0.4553
Range	2435517
Minimum	6519001
Maximum	8954518
Sum	160699624
Count	20

Appendix C

Correlation Coefficient

0.1281

SUMMARY
OUTPUT

Regression Statistics	
Multiple R	0.128081785
R Square	0.016404944
Adjusted R Square	-0.03823923
Standard Error	2804.000941
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	2360408.762	2360408.762	0.300213977	0.590479075
Residual	18	141523583	7862421.277		
Total	19	143883991.8			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	19019.09748	6756.718154	2.814842509	0.011465953	4823.759395	33214.43557
X Variable 1	0.000458763	0.000837284	0.547917856	0.590479075	-0.001300306	0.002217832